

HDOC day-40

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Batch- 15

1. Algorithm

1. Start.
2. Read the student's first name, middle name, and last name.
3. Display the menu repeatedly until the user chooses Exit.
4. According to the menu choice:
 - **1:** Combine first, middle, and last names using `strcpy()` and `strcat()`.
 - **2:** Form the full name and find its length using `strlen()`.
 - **3:** Copy the full name to another string using `strcpy()`.
 - **4:** Compare first and last names using `strcmp()`.
 - **5:** Create the name in reverse order: last, middle, first.
 - **6:** Count vowels, consonants, digits, and spaces using functions from `ctype.h`.
 - **7:** Read a word and search for it in the full name using `strstr()`.
 - **8:** Exit.
5. For an invalid menu choice, display an error message.
6. Stop.

2. Corrected Function Prototypes

The prototypes in the question such as `findLength(..., size_t length)` do not actually return the result to `main()`. To satisfy the **call-by-reference** requirement, use pointers:

```
void combineNames(const char firstName[], const char
middleName[],
                const char lastName[], char fullName[]);
```

```
void findLength(const char fullName[], size_t *length);
```

```
void copyName(const char source[], char destination[]);
```

```
void compareNames(const char firstName[], const char
lastName[],
                int *comparisonResult);
```

```

void reverseNameOrder(const char firstName[], const char
middleName[],
                    const char lastName[], char
reverseName[]);

void countCharacters(const char fullName[], int *vowels,
                    int *consonants, int *digits, int
*spaces);

void searchWord(const char fullName[], const char word[],
                int *searchResult);

```

3. Complete C Program

```

#include <stdio.h>

#include <string.h>

#include <ctype.h>

/* Function Prototypes */

void combineNames(const char firstName[], const char middleName[],
                const char lastName[], char fullName[]);

void findLength(const char fullName[], size_t *length);

void copyName(const char source[], char destination[]);

void compareNames(const char firstName[], const char lastName[],
                int *comparisonResult);

void reverseNameOrder(const char firstName[], const char middleName[],
                const char lastName[], char reverseName[]);

void countCharacters(const char fullName[], int *vowels,
                int *consonants, int *digits, int *spaces);

void searchWord(const char fullName[], const char word[],

```

```

        int *searchResult);

/* Combine first, middle and last names */

void combineNames(const char firstName[], const char middleName[],

        const char lastName[], char fullName[])

{

    strcpy(fullName, firstName);

    strcat(fullName, " ");

    strcat(fullName, middleName);

    strcat(fullName, " ");

    strcat(fullName, lastName);

}

/* Find length of full name */

void findLength(const char fullName[], size_t *length)

{

    *length = strlen(fullName);

}

/* Copy full name */

void copyName(const char source[], char destination[])

{

    strcpy(destination, source);

}

/* Compare first name and last name */

void compareNames(const char firstName[], const char lastName[],

```

```

        int *comparisonResult)

```

```

{
    *comparisonResult = strcmp(firstName, lastName);
}

/* Reverse name order */

void reverseNameOrder(const char firstName[], const char middleName[],
                      const char lastName[], char reverseName[])
{
    strcpy(reverseName, lastName);

    strcat(reverseName, " ");

    strcat(reverseName, middleName);

    strcat(reverseName, " ");

    strcat(reverseName, firstName);
}

/* Count vowels, consonants, digits and spaces */

void countCharacters(const char fullName[], int *vowels,
                    int *consonants, int *digits, int *spaces)
{
    int i;

    char ch;

    *vowels = 0;

    *consonants = 0;

    *digits = 0;

    *spaces = 0;

    for (i = 0; i < strlen(fullName); i++)

```

```

{
    ch = tolower((unsigned char)fullName[i]);

    if (isalpha((unsigned char)ch))
    {
        if (ch == 'a' || ch == 'e' || ch == 'i' ||
            ch == 'o' || ch == 'u')
        {
            (*vowels)++;
        }
        else
        {
            (*consonants)++;
        }
    }
    else if (isdigit((unsigned char)ch))
    {
        (*digits)++;
    }
    else if (isspace((unsigned char)ch))
    {
        (*spaces)++;
    }
}

```

```

/* Search for a word in the full name */

void searchWord(const char fullName[], const char word[],

               int *searchResult)

{
    if (strstr(fullName, word) != NULL)

        *searchResult = 1;

    else

        *searchResult = 0;
}

/* Main Function */

int main(void)

{
    char firstName[50];

    char middleName[50];

    char lastName[50];

    char fullName[160];

    char copiedName[160];

    char reverseName[160];

    char word[50];

    int choice;

    int comparisonResult;

    int vowels, consonants, digits, spaces;

    int searchResult;

    size_t length;

```

```

printf("Enter first name: ");

scanf("%49s", firstName);

printf("Enter middle name: ");

scanf("%49s", middleName);

printf("Enter last name: ");

scanf("%49s", lastName);

/* Form full name initially */

combineNames(firstName, middleName, lastName, fullName);

do

{

    printf("\n===== MENU =====\n");

    printf("1. Combine names and display full name\n");

    printf("2. Find length of full name\n");

    printf("3. Copy full name\n");

    printf("4. Compare first name and last name\n");

    printf("5. Display name in reverse order\n");

    printf("6. Count vowels, consonants, digits and spaces\n");

    printf("7. Search a word in full name\n");

    printf("8. Exit\n");

    printf("===== \n");

    printf("Enter your choice: ");

    scanf("%d", &choice);

    switch (choice)

    {

```

case 1:

```
combineNames(firstName, middleName,
```

```
lastName, fullName);
```

```
printf("Full name: %s\n", fullName);
```

```
break;
```

case 2:

```
findLength(fullName, &length);
```

```
printf("Length of full name = %zu\n", length);
```

```
break;
```

case 3:

```
copyName(fullName, copiedName);
```

```
printf("Copied name: %s\n", copiedName);
```

```
break;
```

case 4:

```
compareNames(firstName, lastName,
```

```
&comparisonResult);
```

```
if (comparisonResult == 0)
```

```
printf("First name and last name are equal.\n");
```

```
else
```

```
printf("First name and last name are different.\n");
```

```
break;
```

case 5:

```
reverseNameOrder(firstName, middleName,
```

```

        lastName, reverseName);

printf("Name in reverse order: %s\n",

        reverseName);

break;

case 6:

    countCharacters(fullName, &vowels,

        &consonants, &digits, &spaces);

printf("Number of vowels    = %d\n", vowels);

printf("Number of consonants = %d\n", consonants);

printf("Number of digits    = %d\n", digits);

printf("Number of spaces    = %d\n", spaces);

break;

case 7:

    printf("Enter word to search: ");

    scanf("%49s", word);

searchWord(fullName, word, &searchResult);

if (searchResult == 1)

    printf("\n\"%s\" occurs in the full name.\n",

        word);

else

    printf("\n\"%s\" does not occur in the full name.\n",

        word);

break;

```

case 8:

```
printf("Exiting program...\n");
```

```
break;
```

default:

```
printf("Invalid choice! Please enter 1 to 8.\n");
```

```
}
```

```
} while (choice != 8);
```

```
return 0;
```

```
}
```

4. Test Case Table

Test	Input / Operation	Expected Result
1	First=Aaryan, Middle=Pratap, Last=Singh, Choice 1	Aaryan Pratap Singh
2	Same names, Choice 2	Length = 18
3	Same names, Choice 3	Copied name = Aaryan Pratap
4	First=Aaryan, Last=Singh, Choice 4	Names are different
5	First=Singh, Last=Singh, Choice 4	Names are equal
6	First=Aaryan, Middle=Pratap, Last=Singh, Choice 5	Singh Pratap Aaryan
7	Aaryan Pratap Singh, Choice 6	Vowels = 5, Consonants = 11, Digits = 0, Spaces = 2
8	Search Pratap	Word occurs in full name
9	Search Kumar	Word does not occur
10	Choice = 10	Invalid choice message
11	Choice = 8	Program exits

EXECUTION

```
[aaryanrajput@aaryans-macbook ~ % nano day40.c
[aaryanrajput@aaryans-macbook ~ % clang day40.c -o day40
[aaryanrajput@aaryans-macbook ~ % ./day40
Enter first name: AARYAN
Enter middle name: PRATAP
Enter last name: SINGH

===== MENU =====
1. Combine names and display full name
2. Find length of full name
3. Copy full name
4. Compare first name and last name
5. Display name in reverse order
6. Count vowels, consonants, digits and spaces
7. Search a word in full name
8. Exit
=====
Enter your choice: 1
Full name: AARYAN PRATAP SINGH

===== MENU =====
1. Combine names and display full name
2. Find length of full name
3. Copy full name
4. Compare first name and last name
5. Display name in reverse order
6. Count vowels, consonants, digits and spaces
7. Search a word in full name
8. Exit
=====
Enter your choice: 2
Length of full name = 19

===== MENU =====
1. Combine names and display full name
2. Find length of full name
3. Copy full name
4. Compare first name and last name
5. Display name in reverse order
6. Count vowels, consonants, digits and spaces
7. Search a word in full name
8. Exit
=====
Enter your choice: 3
Copied name: AARYAN PRATAP SINGH

===== MENU =====
1. Combine names and display full name
2. Find length of full name
3. Copy full name
4. Compare first name and last name
5. Display name in reverse order
6. Count vowels, consonants, digits and spaces
7. Search a word in full name
8. Exit
=====
Enter your choice: 4
First name and last name are different.
```

```
===== MENU =====
1. Combine names and display full name
2. Find length of full name
3. Copy full name
4. Compare first name and last name
5. Display name in reverse order
6. Count vowels, consonants, digits and spaces
7. Search a word in full name
8. Exit
=====
Enter your choice: 5
Name in reverse order: SINGH PRATAP AARYAN

===== MENU =====
1. Combine names and display full name
2. Find length of full name
3. Copy full name
4. Compare first name and last name
5. Display name in reverse order
6. Count vowels, consonants, digits and spaces
7. Search a word in full name
8. Exit
=====
Enter your choice: 6
Number of vowels      = 6
Number of consonants = 11
Number of digits     = 0
Number of spaces      = 2

===== MENU =====
1. Combine names and display full name
2. Find length of full name
3. Copy full name
4. Compare first name and last name
5. Display name in reverse order
6. Count vowels, consonants, digits and spaces
7. Search a word in full name
8. Exit
=====
Enter your choice: 7
Enter word to search: P
"P" occurs in the full name.

===== MENU =====
1. Combine names and display full name
2. Find length of full name
3. Copy full name
4. Compare first name and last name
5. Display name in reverse order
6. Count vowels, consonants, digits and spaces
7. Search a word in full name
8. Exit
=====
Enter your choice: 8
Exiting program...
aaryanrajput@aaryans-macbook ~ %
```